AUTOMATION AND ROBOTICS CLUB
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BUZZER SYSTEM

User Manual

A complete, step-by-step guide for the user.

Connect
Join the ESP32 Wi-Fi network before opening the display.

Display
See a clear green connection status when ready.

Reset
Press Enter after each buzzer result.

Hardware: ESP32 buzzer Software: Buzzer Display application

Welcome To Buzzer System

What it is and how the two parts fit together

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What is the Buzzer System?

The Buzzer System is an ESP32-based buzzer setup used during a quiz or buzzer round. When a participant presses their buzzer, their number appears in large yellow text on the display, so everyone can see who pressed.

This manual explains how to connect and operate the Buzzer Display. It is intended for the person running the display during a quiz or buzzer round.

Part 1 **Hardware**

The ESP32 buzzer unit that the participants use. It is powered on first and provides the Wi-Fi network.

Part 2 **Software**

The Buzzer Display application, which connects to the ESP32 and shows the number of the participant who pressed.

Quick reference

Item	Default or action
ESP32 Wi-Fi password	**12345678**
Default ESP32 address	192.168.4.1
TCP port	80
Reset display	Press **Enter**
Leave full-screen	Press **Esc**

Getting connected

1. Power on the ESP32, then connect the display computer to its Wi-Fi network using the password above.
2. Open the Buzzer Display application. Check that the IP is **192.168.4.1** and the port is **80**.
3. Select **Connect**.
4. The connection is ready when the status turns **green** and the button changes to **Disconnect**.

! **If connection fails.** Confirm that the computer is still on the ESP32 Wi-Fi, then recheck the IP address and port before selecting **Connect** again.

Operating The Display 2

How to setup Hardware - Buzzer?

The hardware consists of 6 buzzers connected to an ESP32 box with 6 connectors for the buzzer wires. Connect each buzzer to its corresponding connector. Power the ESP32 using a USB cable connected to a power adapter or a laptop. If using a laptop, ensure it has sufficient battery charge. The ESP32 creates its own Wi-Fi network, which the display computer connects to in order to receive buzzer presses.

Operator controls

Control	What it does
Enter	Clears the current buzzer number and returns the display to Waiting.
Esc	Leaves full-screen mode. The application remains open.
Disconnect	Closes the connection and changes the button back to Connect.

Quick troubleshooting

Problem	Action
Connection fails	Reconnect to ESP32 Wi-Fi & confirm **192.168.4.1** with port **80**.
Status changes to Disconnected	Check ESP32 power and Wi-Fi, then select **Connect** again.
Number will not change	Press **Enter** to reset the current result before the next round.

Event day checklist

1. Power on the ESP32 and the display computer.
2. Connect the computer to the ESP32 Wi-Fi network.
3. Open the display and select **Connect**.
4. Verify the green connected status before starting the round.
5. The number resets after **10 seconds**, then displays **Waiting...** for next round..

> i **Remember:** If the number gets stuck after the buzzer result, press **Enter.** The display shows ***Waiting...*** again indicating that the system is ready for the next round.

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Connection status

The application shows the connection state at the bottom of the screen.



Fig 1 : Before connecting
The red status reads **Disconnected**.

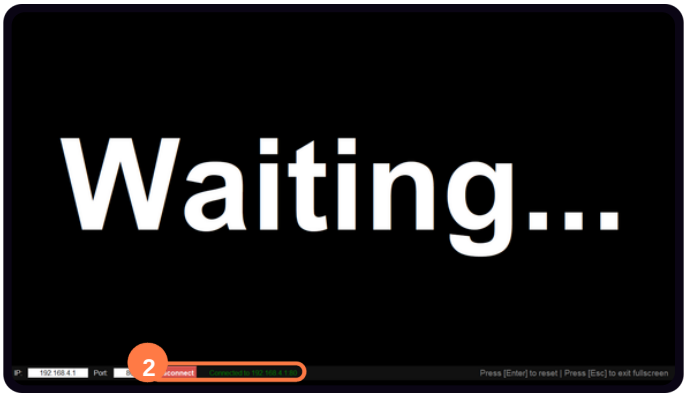


Fig 2 : After connecting
The green status confirms the display is **connected**.

Showing a buzzer result

When a participant presses a buzzer, their number appears in large yellow text. The result remains on screen for **10 seconds** until the next reset .

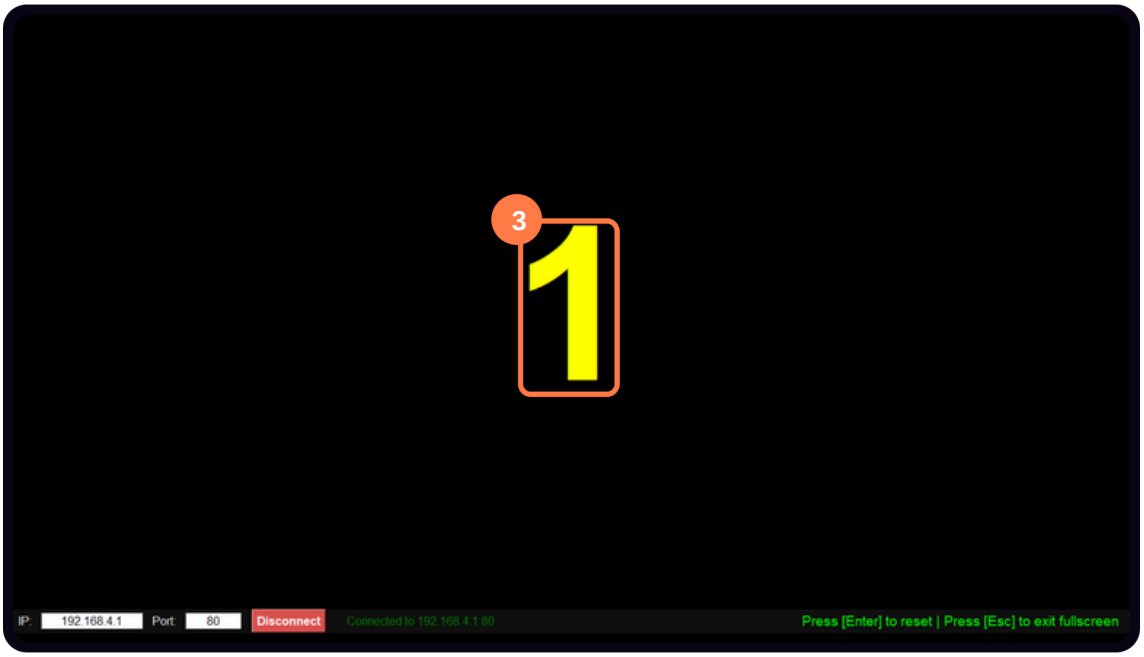


Fig 3 :A participant's number shown in large yellow text, here participant 1.